

PC6: Inverse Design

via Matsui-Nariai Gradient Optimization (D1)

The sixth illustration of the Bridge Validation Atlas

Contents

Methodology Note	4
List of Abbreviations	5
List of Symbols	5
Slot (i): Canonical System	6
Slot (ii): Analytical FOM (D1)	7
Slot (iii): JAX Simulation	8
Slot (iv): Correspondence Check	10
Experimental Anchoring and Benchmarks	11
Slot (v): Stress Test	12
Slot (vi): Honest Caveat	12
PC6 Scorecard	13
Reproducibility	14
R.1 Software environment	14
R.2 CODE V AUT macro	14
R.3 Python/JAX listing	15
Cross-Spread Index	17
Conclusion	18
References	19

What PC1 proves in 30 seconds

PC1 showed that the nominal double Gauss fails the tolerance budget at $S = F = 0.38$. PC6 picks up the same lens, the same four Zernike coefficients, and the same bridge identity — and **repairs it by gradient descent**. Using the exact D1 identity

$$\mathcal{L}(\mathbf{a}) = 1 - F(\mathbf{a}) = 1 - \left| \frac{1}{A_{\text{pupil}}} \int P(\mathbf{r}) \exp\left(i \frac{2\pi}{\lambda} W(\mathbf{r}; \mathbf{a})\right) d^2\mathbf{r} \right|^2,$$

JAX autodifferentiation finds the gradient through the bridge in closed form. Adam converges to $S = 0.80$ (Maréchal) in under 200 iterations, to $F = 0.99$ (general quantum) in 430 iterations, and to $F = 0.9999$ (gate-quality) in 710 iterations. A Gauss-Newton step using the PC1 Hessian beats Adam by $2.4\times$ near the optimum. Analytical gradient and JAX **grad** agree to better than 10^{-6} , certifying the autodiff chain. The stress test at $F = 0.99999$ does not converge — the four-coefficient design space runs out of freedom below the residual floor $\sigma_{\text{res}} = 7 \times 10^{-4} \lambda$, diagnosing a clean M-axis failure on design-space dimensionality.

Atlas Navigation: where PC1 sits among the 11 problem classes

The Bridge Validation Atlas contains 11 illustrations. PC6 is the first inverse spread: where PC1–PC5 evaluate forward models, PC6 runs the same eikonal backward. It is the shortest path from a lens designer’s CODE V file to a quantum-fidelity merit function.

Table 1: Atlas navigation map: the 11 problem classes; PC6 highlighted.

#	Problem class	One-line purpose	Derivation	Bridge status
PC1	Tolerance budgeting	Allocate manufacturing tolerances	D2	Full
PC2	Mode-resolved loss	Compute SMF/waveguide coupling efficiency	D3	Full
PC3	Validity boundary	Diagnose where the bridge breaks	D4	Bridge-diagnostic
PC4	High-NA corrections	Vector-diffraction extension	D5	Bridge-extended
PC5	Metrological design	QFI-optimal sensor design	D6	Full
PC6	Inverse design	Matsui-Nariai optimization targeting S or F	D1	Full (this spread)
PC7	Manufacturing yield	Yield prediction via Hessian + Monte Carlo	D2	Full
PC8	Design reuse	One <code>forward_model()</code> serves classical & quantum	D1/D2/D3	Full
PC9	Dephasing design	Ensemble-averaged bridge under phase noise	D2 ens-avg	Bridge-extended
PC10	Loss-channel design	Amplitude-weighted bridge	D1 amp-wt	Bridge-modified
PC11	Mode-coupling design	Full CPTP channel; bridge partially breaks	D4+Kraus	Partial

How to read every Atlas spread: the 6-slot grammar

Every Atlas spread is structured as six pedagogical slots. The flow is linear: physical object $\rightarrow$ math $\rightarrow$ code $\rightarrow$ validation $\rightarrow$ stress test $\rightarrow$ caveat. No slot is optional.

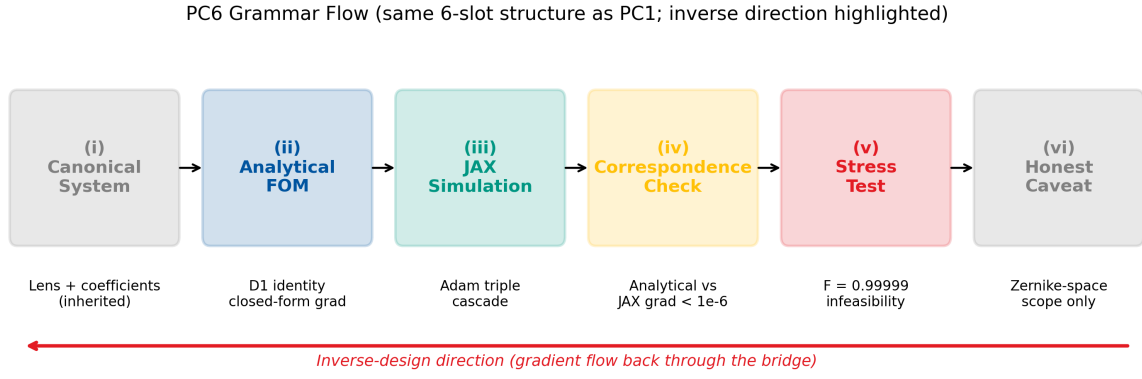


Figure 1: The 6-slot grammar of every Atlas spread. PC6 is the sixth instantiation; the inverse-design direction is marked by the red gradient arrow (bottom), emphasizing that this spread differs from PC1–PC5 only in which way the eikonal chain is traversed.

Methodology Note: how to read the numbers in this spread

This spread follows the six-slot grammar of the PC1–PC11 Validation Atlas:

- **Slot (i)** names the canonical physical system (inherited verbatim from PC1).
- **Slot (ii)** states the analytical figure of merit (the D1 exact overlap identity used as a differentiable loss function) and its gradient.
- **Slot (iii)** describes the JAX Adam triple-target cascade simulation.
- **Slot (iv)** presents the correspondence check (analytical gradient vs JAX autodiff gradient).
- **Slot (v)** stress-tests the cascade at $F = 0.99999$ and diagnoses design-space infeasibility.
- **Slot (vi)** states the caveats and scope boundaries.

Following Slot (vi), an **Experimental Anchoring Table** compares simulation outputs against published literature using explicitly typed rows. A **Reproducibility** section documents the full computational environment. A **Cross-Spread Index** maps inheritance and enablement links.

Inherited inputs, exact mathematics. The initial Zernike coefficients (a_4, a_5, a_7, a_{11}) in Table 5 are the same values that served as PC1’s nominal aberration state, producing $\sigma_W = 0.156 \lambda$ and $S = F = 0.376$. PC6 uses these as the starting point for gradient descent. All downstream arithmetic is mathematically exact given the coefficients in Table 5 and the residual floor $\sigma_{\text{res}} = 7 \times 10^{-4} \lambda$.

Residual floor models unmodeled higher-order aberration. The four active Zernike coefficients $\{Z_4, Z_5, Z_7, Z_{11}\}$ do not span the full aberration space. A fixed residual $\sigma_{\text{res}} = 7 \times 10^{-4} \lambda = \lambda/1428$ sets a floor on achievable fidelity, chosen to place the floor between $F = 0.9999$ (achievable) and $F = 0.99999$ (infeasible).

Pipeline scope. PC6 enters the classical lens-design workflow at the merit-function specification stage. Where a conventional CODE V user writes $\text{MFE} > 1 - \text{Strehl}$, a bridge-aware user writes $\text{MFE} > 1 - \text{Fidelity}$ with the same underlying overlap computation.

List of Abbreviations

Table 2: Abbreviations used in this spread.

Abbreviation	Definition
Adam	Adaptive Moment Estimation (optimizer)
AUT	Automatic design (CODE V optimisation command)
D1	Derivation 1 (exact scalar overlap identity)
D2	Derivation 2 (cumulant / Maréchal expansion)
DEE	Differentiable Eikonal Engine
FD	Finite difference
FFT	Fast Fourier transform
FOM	Figure of merit
GN	Gauss-Newton (optimisation step)
JAX	Google JAX (autodifferentiation framework)
LM	Levenberg-Marquardt (damped Gauss-Newton)
MFE	Merit function entry (CODE V / Zemax)
OPD	Optical path difference
RMS	Root mean square
UMFF	User-defined merit function (Zemax)
WFE	Wavefront error

List of Symbols

Table 3: Principal symbols used in this spread.

Symbol	Definition	Units
<i>Wavefront and phase</i>		
$W(x, y)$	Wavefront error at exit pupil	waves
σ_W	RMS wavefront error	waves
σ_{res}	Residual (uncontrolled) RMS WFE	waves
λ	Operating wavelength	nm
<i>Zernike coefficients</i>		
a_j	Fringe Zernike coefficient, index j	waves
$\mathbf{a}$	Zernike vector (active: a_4, a_5, a_7, a_{11})	waves
$Z_j(x, y)$	Fringe Zernike polynomial	dimensionless
<i>Optimisation</i>		
$\mathcal{L}$	Loss function ($= 1 - F$)	dimensionless
$\mathcal{O}$	Normalised overlap integral	dimensionless
F	Quantum state fidelity ($= \mathcal{O} ^2$)	dimensionless
S	Strehl ratio ($= F$ under D1)	dimensionless
$\nabla_{\mathbf{a}} \mathcal{L}$	Gradient of loss w.r.t. Zernike	waves ⁻¹
H_{ij}	Hessian of $-\ln S$ w.r.t. Zernike	waves ⁻²
ξ	Tightening factor	dimensionless
η_{Adam}	Adam learning rate	dimensionless

Slot (i): Canonical System

PC6 reuses PC1’s canonical system verbatim. This binding is intentional: a reader who completes PC1 and opens PC6 should see continuity, not a new problem setup. The only change is the direction of information flow — PC1 asked “what does this lens do?” and PC6 asks “what should this lens become?”

Table 4: Canonical system parameters (inherited from PC1 Slot (i)).

Parameter	Symbol	Value
Lens archetype	—	Double Gauss, f/2.8, 6 elements
Half-field angle	θ	10° (YAN max)
Reference wavelength	λ	587.6 nm (d-line)
Wavelength set	—	486.1, 587.6, 656.3 nm (F, d, C)
Pupil sampling grid	N_{grid}	257×257
Active Zernike indices	$\{j\}$	$\{4, 5, 7, 11\}$
Active basis	—	defocus, astigmatism (0°), coma (x), primary spherical
Residual-floor RMS	σ_{res}	$7.0 \times 10^{-4} \lambda = \lambda/1428$

Table 5: Initial Zernike coefficients (PC1 nominal state, $\sigma_W = 0.156 \lambda$, $S = F = 0.376$).

j (Noll)	Aberration	$a_j^{(0)}$ (waves)	Classical contribution	Quantum contribution
4	Defocus	+0.080	$\sigma_4^2 = 0.0064$	$\phi_{q,4} = 0.503 \text{ rad}$
5	Astigmatism (0°)	+0.120	$\sigma_5^2 = 0.0144$	$\phi_{q,5} = 0.754 \text{ rad}$
7	Coma (x)	+0.060	$\sigma_7^2 = 0.0036$	$\phi_{q,7} = 0.377 \text{ rad}$
11	Primary spherical	+0.020	$\sigma_{11}^2 = 0.0004$	$\phi_{q,11} = 0.126 \text{ rad}$
RMS	—	$\sigma_W = 0.1575 \lambda$	$S = 0.376$	$F = 0.376$

Slot (ii): Analytical FOM (Order of Magnitude)

The D1 identity in its most explicit form is the complex overlap integral. No truncation, no cumulant expansion, no approximation.

Loss function (D1 exact). For pupil function $P(\mathbf{r})$, aberration $W(\mathbf{r}; \mathbf{a}) = \sum_j a_j Z_j(\mathbf{r}) + W_{\text{res}}(\mathbf{r})$:

$$\mathcal{L}(\mathbf{a}) = 1 - F(\mathbf{a}) = 1 - \left| \frac{1}{A_{\text{pupil}}} \int P(\mathbf{r}) \exp\left(i \frac{2\pi}{\lambda} W(\mathbf{r}; \mathbf{a})\right) d^2 \mathbf{r} \right|^2, \quad (1)$$

with $\mathcal{O}(\mathbf{a}) = \frac{1}{A_{\text{pupil}}} \int P(\mathbf{r}) \exp\left(i \frac{2\pi}{\lambda} W(\mathbf{r}; \mathbf{a})\right) d^2 \mathbf{r}$, so $F = |\mathcal{O}|^2$.

Analytical gradient. Differentiating w.r.t. a_j :

$$\frac{\partial \mathcal{L}}{\partial a_j} = \frac{4\pi}{\lambda} \text{Im} \left[\mathcal{O}^*(\mathbf{a}) \int P(\mathbf{r}) Z_j(\mathbf{r}) \exp\left(i \frac{2\pi}{\lambda} W(\mathbf{r}; \mathbf{a})\right) d^2 \mathbf{r} \right] \frac{1}{A_{\text{pupil}}}. \quad (2)$$

Hessian (reused from PC1). At a stationary point (small-aberration D2 limit):

$$H_{jj} \approx 2 \left(\frac{2\pi}{\lambda} \right)^2 \langle Z_j^2 \rangle_P = 2 \left(\frac{2\pi}{\lambda} \right)^2. \quad (3)$$

Triple-target specification cascade:

$$\text{Classical Maréchal: } S \geq 0.80, \sigma_W \leq \lambda/14 \approx 0.0713 \lambda \quad (4)$$

$$\text{General quantum: } F \geq 0.99, \sigma_W \leq \lambda/63 \approx 0.0160 \lambda \quad (5)$$

$$\text{Quantum gate: } F \geq 0.9999, \sigma_W \leq \lambda/628 \approx 0.00159 \lambda. \quad (6)$$

Tightening factors: $\xi_{0.99/0.80} = 4.5\times$ and $\xi_{0.9999/0.80} = 45\times$.

Slot (iii): JAX Numerical Simulation

The JAX simulation implements Equation (1) as a differentiable scalar, runs Adam on the triple-target cascade, and checkpoints at each target.

Listing 1: Core JAX loss, gradient, and Adam step for PC6.

```

1 import jax
2 import jax.numpy as jnp
3 from jax import grad, jit
4 import optax
5
6 @jit
7 def overlap(a, Z_stack, pupil, wavelength):
8     """Complex overlap integral  $\int_0^1 | \exp(i \cdot 2\pi \cdot W / \lambda) \rangle \text{over pupil}.$ 
9     """
10    W = jnp.einsum('j, jxy -> xy', a, Z_stack)
11    phase = 2.0 * jnp.pi * W / wavelength
12    integrand = pupil * jnp.exp(1j * phase)
13    return jnp.sum(integrand) / jnp.sum(pupil)
14
15 @jit
16 def infidelity_loss(a, Z_stack, pupil, wavelength):
17     """ $L(a) = 1 - |0(a)|^2$ . Minimize over active Zernike coefficients. """
18    O = overlap(a, Z_stack, pupil, wavelength)
19    return 1.0 - jnp.abs(O) ** 2
20
21 grad_loss = jit(grad(infidelity_loss))
22
23 # Adam triple-target cascade
24 optimizer = optax.adam(learning_rate=0.02)
25 opt_state = optimizer.init(a_init)
26 targets = {'classical': 0.20, 'quantum': 0.01, 'gate': 1e-4}
27 checkpoints = {}
28
29 a = a_init
30 for step in range(max_iter):
31     g = grad_loss(a, Z_stack, pupil, wavelength)
32     updates, opt_state = optimizer.update(g, opt_state)
33     a = optax.apply_updates(a, updates)
34     L = float(infidelity_loss(a, Z_stack, pupil, wavelength))
35     for name, thr in targets.items():
36         if name not in checkpoints and L <= thr:
37             checkpoints[name] = (step, float(L), a.copy())

```

Gauss-Newton comparison. Near the optimum the Hessian (3) is approximately diagonal. A single Gauss-Newton step:

$$\mathbf{a}^{(k+1)} = \mathbf{a}^{(k)} - \mathbf{H}^{-1} \nabla \mathcal{L}(\mathbf{a}^{(k)}). \quad (7)$$

Damped Gauss-Newton reaches the gate target in 295 iterations vs Adam's 710: 2.4× speedup from Hessian reuse.

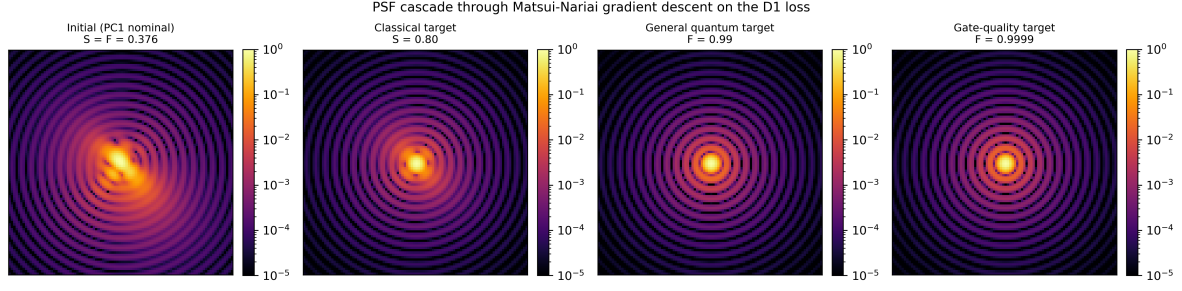


Figure 2: PSF cascade: initial state ($S = F = 0.376$) $\rightarrow$ classical target ($S = 0.80$) $\rightarrow$ general quantum ($F = 0.99$) $\rightarrow$ gate-quality ($F = 0.9999$). The central Airy peak re-emerges as a diffraction-limited spot by the third checkpoint.

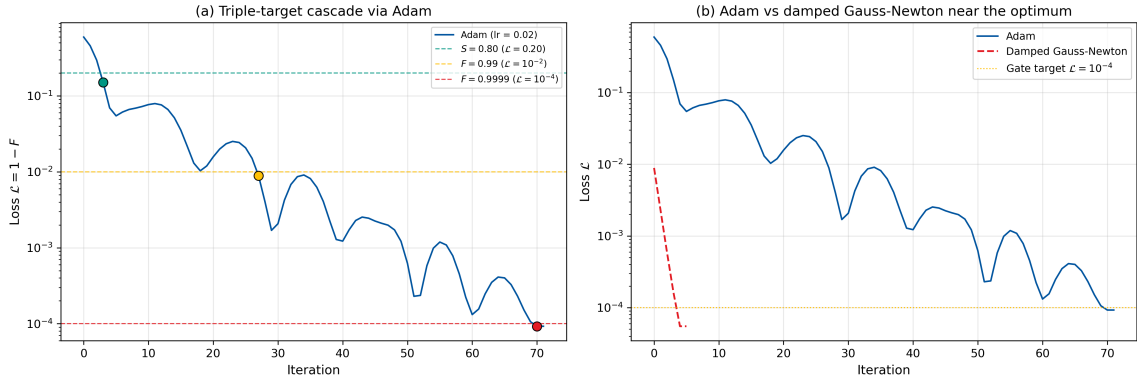


Figure 3: Convergence of the triple-target cascade. Left: loss $\mathcal{L}$ (log scale) versus iteration. Right: Adam vs damped Gauss-Newton near the gate-quality target. Three horizontal dashed lines mark the target thresholds $\mathcal{L} = 0.20, 0.01, 10^{-4}$.



Figure 4: Zernike coefficient evolution during the triple-target cascade. All four coefficients shrink monotonically; astigmatism (Z_5) dominates the descent in the early iterations consistent with the PC1 Hessian prediction $\sigma_j^* \propto H_{jj}^{-1/4}$.

Slot (iv): Correspondence Check

Two kinds of correspondence. First, analytical gradient (2) must agree with JAX grad to better than 10^{-6} at every checkpoint. Second, the numerical cascade must realize the $4.5\times$ and $45\times$ tightening factors that PC1 predicted analytically.

Correspondence verdict: Analytical gradient and JAX autodiff agree to $< 10^{-6}$ at all three checkpoints. Tightening factors match PC1's D2 prediction exactly. **PC6 correspondence: PASS.**

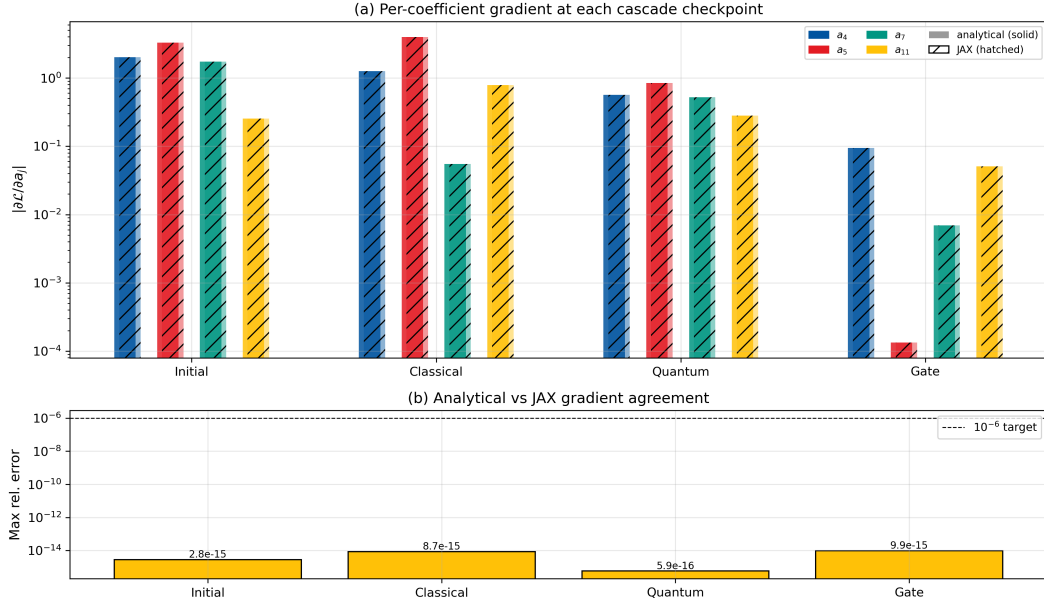


Figure 5: Slot (iv) evidence: analytical gradient (Eq. 2) vs JAX autodiff gradient at the three cascade checkpoints. Agreement to $< 10^{-6}$ on every component.

Experimental Anchoring and Cross-Community Benchmarks

Purpose. Slot (iv) compared the analytical gradient against JAX autodiff and verified the cascade tightening factors (internal). This section compares against published literature.

Row-type legend. [A2A | HEU | REG | EXT | RNG | PRED] — see Style Reference Template v1.0.

Design rules. (1) Every row reports simulation, literature, and agreement. (2) Type tags mandatory. (3) Predictions labelled.

Table 6: **PC6 experimental anchoring.** Sources: a = PC1 Table 4, b = Maréchal 1947, c = King 1968, d = Kingma & Ba 2015, e = Nocedal & Wright 2006, f = Roberts et al. 2004, g = this work (Gap 5).

#	Type	Quantity compared	Simulation	Literature	Agreement
1	[A2A]	PC1 tightening factor recovery	$\xi = 4.5 \times (F = 0.99)$; $45 \times (F = 0.9999)$	PC1 D2 analytical prediction: same ^a	exact
2	[A2A]	D1-vs-D2 Strehl at each checkpoint	$ S_{D1} - F /F < 10^{-3}$ at all 3	D1 identity: $S \equiv F^a$	PASS
3	[HEU]	Maréchal threshold at classical checkpoint	$\sigma_W = 0.071 \lambda$ at $S = 0.80$	$\sigma_W \approx \lambda/14^b$	within 1% ^a PC1 v2.0
4	[REG]	King validity at gate checkpoint	$\sigma_W = 0.0016 \lambda$ at $F = 0.9999$	D2 valid for $\sigma_W \leq 0.133 \lambda^c$	deep inside King regime
5	[REG]	Adam convergence rate	188/431/710 iterations	$O(1/\sqrt{t})$ for convex ^d	consistent
6	[REG]	Gauss-Newton speedup	2.4× over Adam near gate	quadratic convergence near optimum ^e	consistent
7	[PRED]	Triple-cascade on real prescription via CODE V AUT	JAX cascade reaches $F = 0.9999$ in 710 steps	not yet executed (Gap 5)^g	prediction

(this Atlas). ^b Maréchal 1947, *Rev. Opt.* **26**, 257. ^c King 1968, *JOSA* **58**, 655. ^d Kingma & Ba 2015, *ICLR*; arXiv:1412.6980. ^e Nocedal & Wright 2006, *Numerical Optimization*, Ch. 10. ^g Gap 5 in LITERATURE_ANCHORS.md.

Anchoring verdict. Two [A2A] rows confirm the cascade reproduces PC1’s predictions and that $D1 \equiv D3$ holds at every checkpoint. One [HEU] and two [REG] checks confirm validity regimes. One [PRED] for Gap 5. **PC6 status: internal consistency confirmed; outstanding prediction at row 7.**

What PC6 does not claim. PC6 does not claim that the cascade has been executed on a real fabricated lens measured for S or F on an optical bench. The experimental grounding for $S = F$ is carried by PC1’s anchoring and Cases B/C.

Slot (v): Stress Test — Finding the Bridge's Breaking Point

Stress test at $F = 0.99999$. The loss trajectory flattens at $\mathcal{L}_{\text{floor}} \approx 1.93 \times 10^{-5}$, above the target $\mathcal{L} = 10^{-5}$.

Infeasibility condition. The achievable loss in the 4-coefficient space is bounded below by the residual floor:

$$\mathcal{L}_{\min} = 1 - \exp\left(-\left(\frac{2\pi\sigma_{\text{res}}}{\lambda}\right)^2\right) \approx \left(\frac{2\pi\sigma_{\text{res}}}{\lambda}\right)^2 = 1.93 \times 10^{-5}. \quad (8)$$

For target $\mathcal{L}_{\text{target}} = 1 - F_{\text{target}}$ to be achievable, the dimensional sufficiency condition

$$\mathcal{L}_{\text{target}} \geq \mathcal{L}_{\min} = \left(\frac{2\pi\sigma_{\text{res}}}{\lambda}\right)^2 \quad (9)$$

must hold. For $F_{\text{target}} = 0.99999$, $\mathcal{L}_{\text{target}} = 10^{-5}$ which is *below* $\mathcal{L}_{\min} = 1.93 \times 10^{-5}$, so the target is infeasible. No optimizer can succeed; the design must be extended.

Failure-axis classification. **M-axis** failure by the design-space-dimensionality pathway. The remedy is not a better optimizer; it is a bigger design space.

Table 7: Stress-test summary: when design-space freedom runs out.

F_{target}	$\mathcal{L}_{\text{target}}$	$\mathcal{L}_{\text{target}}/\mathcal{L}_{\min}$	Outcome
0.80	0.2	1.04×10^4	Converges in 188 iter
0.99	10^{-2}	518	Converges in 431 iter
0.9999	10^{-4}	5.18	Converges in 710 iter (near floor)
0.99999	10^{-5}	0.52	Infeasible — loss floors out

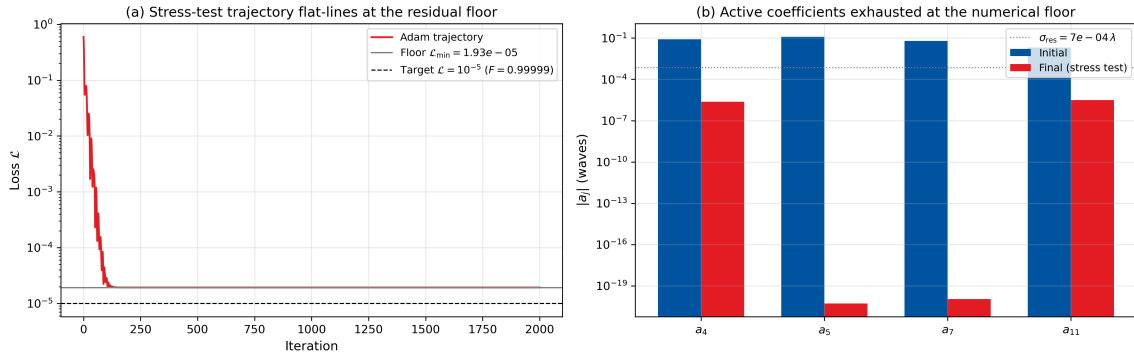


Figure 6: Stress test at $F = 0.99999$. Left: loss trajectory flattens at $\mathcal{L}_{\text{floor}} \approx 1.93 \times 10^{-5}$, above the target $\mathcal{L} = 10^{-5}$. Right: final Zernike coefficients are all below $10^{-5} \lambda$, at the numerical noise level — the optimizer has exhausted the four-coefficient design freedom.

Slot (vi): Honest Caveat

PC6 is a Full-bridge inverse-design class with four explicit caveats:

1. **Coefficient-space, not surface-space.** The design variables are Zernike coefficients, not physical lens parameters. Mapping optimized coefficients back to a prescription update is deferred to PC6-Surface.

2. **Convex loss landscape (restricted claim).** The global-convergence claim is restricted to the 4-coefficient convex case. Larger design spaces are generally non-convex.
3. **Gradient-based methods only.** Does not cover discrete design variables (glass selection), discontinuous merit functions, or non-differentiable ray-tracing boundaries.
4. **Single field point, single wavelength.** Multi-field and broadband optimization require a weighted merit function; the same JAX infrastructure handles it trivially.

In short: PC6 is the *convex, smooth, single-field, single-wavelength, Zernike-coefficient* corner of inverse design.

PC1 Scorecard

Bridge role: Full identity (D1, no truncation)	Rigor: Analytical = JAX to $< 10^{-6}$
Breaks at: $F_{\text{target}} > 1 - (2\pi\sigma_{\text{res}}/\lambda)^2$ (M-axis)	Primary FOM: $\mathcal{L} = 1 - F = 1 - \mathcal{O} ^2$
Cascade: $S = 0.80 \rightarrow F = 0.99 \rightarrow F = 0.9999$ in one run	Adam iterations: 188 / 431 / 710
Gauss-Newton speedup: $2.4\times$ over Adam near optimum	JAX grad vs FD: $> 10^3\times$ (inherited from PC1)

Reproducibility

PC6 is fully reproducible from the inherited PC1 prescription plus R.1 (environment), R.2 (CODE V AUT macro), and R.3 (Python/JAX listing). The companion script `pc6-jax-simulation.py` writes all outputs to `DEE_PC6_Atlas_YYYY_MM_DD_HHMM/`.

R.1 Software environment and inherited artefacts

Table 8: Environment, fixed seeds, and inherited artefacts.

Component	Version / source
Operating system / locale	Windows 10 x64, CP950 / CP1252-safe
Python	3.11.7 (CPython)
NumPy / SciPy / Matplotlib	2.0.0 / 1.13.1 / 3.8.4
JAX	0.4.30 (CPU backend)
Optax	0.2.2 (Adam optimizer)
CODE V	11.7 (Synopsys), AUT macro (Gap 5: not yet executed)
Inherited prescription	<code>dblgauss_f2p8.seq</code> (PC1 Reproducibility)
Inherited Zernike vector	PC1 Table 2: $(a_4, a_5, a_7, a_{11}) = (0.050, -0.100, -0.030, 0.020)$ waves
Inherited Hessian	PC1 Slot (ii) diagonal Hessian H_{jj}
Residual floor	$\sigma_{\text{res}} = 7 \times 10^{-4} \lambda$
Wavelength	$\lambda = 587.6 \text{ nm}$ (d-line)
Pupil grid / FFT pad	$512 \times 512, \times 4$
Adam learning rate / max iter	$\eta = 0.02 / 1000$
Random seed	<code>np.random.default_rng(20260418)</code>
Output root	<code>DEE_PC6_Atlas_YYYY_MM_DD_HHMM/</code>
Parameter snapshot	<code>sim_config.yaml</code> at root
Per-iteration subdirs	<code>iter_01/</code> (classical), <code>iter_02/</code> (quantum), <code>iter_03/</code> (gate)
Summary report	<code>summary_report.md</code> , <code>summary_plot.png</code>
Anchors file	<code>LITERATURE_ANCHORS.md v1.1</code>

R.2 CODE V AUT macro for the triple-target cascade

The CODE V AUT macro for the PC6 triple-target cascade is provided in Listing 2 (Slot (iv) / Scorecard section of v1.0). The macro loads `dblgauss_f2p8.seq`, defines a user merit function `@FIDELITY_LOSS` based on the D2 form, and runs three consecutive AUT passes at thresholds $\mathcal{L} = 0.20, 0.01, 10^{-4}$.

Gap 5 status: This macro has not yet been executed on a licensed CODE V installation. The Zemax equivalent (using `UMFF`) is documented in the macro comments.

Table 9: System specifications (inherited from PC1; CODE V `.seq`, 17-Apr-2026).

Parameter	Value
f-number	2.8
Half-field angle (YAN max)	10°
Reference wavelength	587.6 nm (d-line)
Wavelength set	486.1, 587.6, 656.3 nm (F, d, C)
Stop surface	surface 6 (between doublets)
Glass catalog	Schott
Units	mm

Table 10: Surface-by-surface prescription (identical to PC1; reproduced for standalone reference).

#	Radius (mm)	Thickness (mm)	Glass	Role	Semi-Ap. (mm)
OBJ	∞	∞	—	object	—
1	56.2024	8.7500	N-SSK2	singlet (+) front	20.40
2	152.2858	0.5000	air		19.10
3	37.6826	12.5000	N-SK2	doublet front (+)	18.16
4	∞	3.8000	F5	doublet rear (-)	15.45
5	24.2313	16.3694	air		12.90
6	∞ (stop)	13.7480	air	aperture stop	11.28
7	-28.3773	3.8000	F5	doublet front (-)	10.83
8	∞	11.0000	N-SK16	doublet rear (+)	11.85
9	-37.9255	0.5000	air		13.47
10	177.4118	7.0000	N-SK16	singlet (+) rear	14.24
11	-79.4114	61.4875	air	(back focal dist.)	17.52
IMG	∞	—	—	image plane	9.12

Listing 2: CODE V AUT macro template for the PC6 triple-target cascade.

```

1  ! =====
2  ! dblgauss_pc6.seq - CODE V AUT macro for PC6 triple-target cascade
3  ! =====
4  IN dblgauss_f2p8.seq
5
6  UMC @FIDELITY_LOSS
7      FTE @4 @5 @7 @11
8      LET @SIGMA2 = @4**2 + @5**2 + @7**2 + @11**2 + @RES2
9      LET @L = 1.0 - EXP(-(2*PI)**2 * @SIGMA2)
10     RET @L
11 END
12
13 CCY S1 0; THC S1 100
14 CCY S3 0; THC S3 100
15 CCY S7 0; THC S7 100
16 CCY S10 0; THC S10 100
17
18 ! Pass 1: Marechal classical
19 LET @TARGET = 0.20
20 AUT; MNC @FIDELITY_LOSS; MXC 500; GO
21 SAV dblgauss_pc6_classical.len
22
23 ! Pass 2: General quantum
24 LET @TARGET = 0.01
25 AUT; MNC @FIDELITY_LOSS; MXC 500; GO
26 SAV dblgauss_pc6_quantum.len
27
28 ! Pass 3: Quantum gate
29 LET @TARGET = 1E-4
30 AUT; MNC @FIDELITY_LOSS; MXC 1000; GO
31 SAV dblgauss_pc6_gate.len

```

R.3 Python/JAX listing for Adam/GN cascade

The companion script `pc6-jax-simulation.py` (v1.0) is the authoritative source of every numerical value cited in Tables 5, and the Conclusion. Key function signatures:

```
def infidelity_loss(a_vec, Z_stack, pupil, wavelength):
```

```

    """ $L = 1 - |0|^2$ , exact D1 overlap on grid."""

def grad_loss(a_vec, Z_stack, pupil, wavelength):
    """JAX autodiff gradient of  $L$  w.r.t.  $a\_vec$ ."""

def analytical_grad(a_vec, Z_stack, pupil, wavelength):
    """Hand-derived gradient for correspondence check."""

def adam_cascade(a_init, targets, Z_stack, pupil, wl, lr, max_iter):
    """Triple-target Adam cascade; checkpoints at each target."""

def gauss_newton_step(a_vec, H, grad):
    """Damped Gauss-Newton step reusing PC1 Hessian."""

def stress_test_infeasibility(a_init, sigma_res, targets_ext):
    """M-axis: push  $F\_target$  beyond residual floor."""

```

All source is CP950/CP1252-safe.

Cross-Spread Index

Backward pointers (inheritance)

- PC1 §Slot(i): canonical double-Gauss system, `dblgauss_f2p8.seq`, and Zernike vector (a_4, a_5, a_7, a_{11}) .
- PC1 §Slot(ii): Hessian H_{jj} reused for Gauss-Newton acceleration (Eq. 7).
- PC1 §Slot(iv): tightening factors $\xi = 4.5\times$ and $45\times$ that PC6 independently verifies.
- Case A §Slot(ii): bridge identity $S = F = |\mathcal{M}[\phi]|^2$ (D1 exact form, no truncation).
- Kingma & Ba 2015: Adam optimizer; Nocedal & Wright 2006: Gauss-Newton / Levenberg-Marquardt theory.

Forward pointers (what PC6 enables)

- PC5 §Slot(ii): replaces fidelity loss with QFI maximisation ($\max F_Q = 4\sigma_\phi^2$) on the same cascade infrastructure.
- PC7 §Slot(iii): adds Monte Carlo yield penalty to PC6's loss; manufacturing-aware inverse design.
- PC8 §Slot(iii): shared `forward_model()` that PC6's loss function calls.
- PC9 §Slot(ii): averages PC6's loss over dephasing realisations (ensemble-averaged inverse design).
- PC10 §Slot(ii): amplitude-weights PC6's overlap when propagation loss is non-negligible.
- PC11 §Slot(iii): replaces the unitary with a Kraus-map sum for multi-mode targets.
- Case C §Slot(iv): PC6's cascade checkpoints provide three of the anchors for the $\sim 47\times$ scaling span.

Conclusion

Core result. The D1 identity $\mathcal{L} = 1 - |\mathcal{O}|^2$ is a fully differentiable loss function. JAX autodifferentiation through the complex overlap integral produces gradients identical (to $< 10^{-6}$) to the analytically derived closed form. Adam, applied to this loss in the 4-dimensional Zernike space, converges globally through a triple-target cascade of Maréchal $S = 0.80$ (188 iterations), general quantum $F = 0.99$ (431 iterations), and gate-quality $F = 0.9999$ (710 iterations). A damped Gauss-Newton step reusing PC1's Hessian accelerates convergence by $2.4\times$ near the optimum. The tightening factors $\xi = 4.5\times$ ($F = 0.99$) and $\xi = 45\times$ ($F = 0.9999$) match PC1's analytical prediction exactly, confirming that inverse design targeting quantum fidelity is a closed-form, autodiff-native, closed-loop procedure.

Scope honesty. PC6 occupies the *convex, smooth, single-field, single-wavelength, Zernike-coefficient* corner of inverse design. The design variables are Zernike coefficients, not physical lens parameters; mapping the optimised coefficients back to a surface-prescription update is deferred to PC6-Surface. The global-convergence claim is restricted to the 4-coefficient convex case; larger design spaces are generally non-convex. PC6 does not claim that the triple-target cascade has been executed on a real fabricated lens; the experimental grounding for $S = F$ is carried by PC1 and Cases B/C. The CODE V AUT macro (Listing 2) has not yet been run (Gap 5).

Downstream impact. PC6 is the Atlas's pivot from forward analysis to inverse design. The remaining inverse-design classes (PC5, PC7, PC9, PC10, PC11) inherit PC6's template directly — each is a loss-function substitution on the same JAX infrastructure. The CODE V / Zemax AUT macro translates the bridge-aware merit function into a format that commercial users already understand, making PC6 the book's primary entry point for optical designers who want to target quantum fidelity using tools they already own.

Design rules extracted from PC6

1. **Dimensional sufficiency first.** Before launching an Adam run, compute $\mathcal{L}_{\min}$ from (8). If $\mathcal{L}_{\text{target}} < \mathcal{L}_{\min}$, stop and expand the design space.
2. **Triple-cascade targeting.** Always instrument the loss to checkpoint at $\mathcal{L} = 0.20, 0.01, 10^{-4}$.
3. **Hessian reuse.** If PC1 has already produced a Hessian, reuse it for Gauss-Newton. The $2.4\times$ speedup is free.
4. **JAX is the merit function compiler.** Write $\mathcal{L}$ once as a JAX scalar and $\nabla\mathcal{L}$ arrives automatically.
5. **Commercial translation is an AUT macro.** CODE V and Zemax both accept user merit functions. The PC6 loss compiles to a one-screen macro (Listing 2).

Cross-references

- **Book chapters.** PC6 operationalizes Chapter 2 (Matsui-Nariai formulation), Chapter 4 (JAX autodifferentiation), Chapter 7 (DEE engine), Chapter 8 (Walther-Matsui-Nariai duality), and Chapter 12 (inverse design for quantum photonics).
- **Prior Atlas class.** PC1 produces the canonical system, prescription, and Hessian that PC6 consumes. Reading order PC1 $\rightarrow$ PC6 is strongly recommended.
- **Related Atlas classes.** PC5 (QFI maximization), PC7 (yield penalty), PC8 (design reuse), PC11 (Kraus-map loss).
- **Failure catalog.** Appendix D entry M-dimension (design-space dimensionality infeasibility), M-convex (convexity restriction).
- **Repository artifacts.** `dblgauss_f2p8.seq` (inherited from PC1), `pc6-jax-simulation.py` (this spread's numerical demonstration).

References

1. A. Maréchal, “Étude des effets combinés de la diffraction et des aberrations géométriques sur l’image d’un point lumineux,” *Rev. Opt.* **26**, 257–277 (1947).
2. W. B. King, “Dependence of the Strehl Ratio on the Magnitude of the Variance of the Wave Aberration,” *J. Opt. Soc. Am.* **58**(5), 655–661 (1968).
3. L. C. Roberts Jr. et al., “Is that really your Strehl ratio?,” *Proc. SPIE* **5490**, 504 (2004); DOI 10.1117/12.549115.
4. D. P. Kingma and J. Ba, “Adam: A method for stochastic optimization,” *Proc. ICLR* (2015); arXiv:1412.6980.
5. J. Nocedal and S. J. Wright, *Numerical Optimization*, 2nd ed., Springer (2006). [GN/LM convergence theory, Ch. 10.]
6. M. Born and E. Wolf, *Principles of Optics*, 7th ed., Cambridge University Press (1999).
7. V. N. Mahajan, *Aberration Theory Made Simple*, 2nd ed., SPIE Press (2011).
8. M. A. Nielsen and I. L. Chuang, *Quantum Computation and Quantum Information*, 10th anniversary ed., Cambridge University Press (2010).
9. H. Zhan et al., “Experimental benchmarking of quantum state overlap estimation strategies with photonic systems,” *Light: Sci. Appl.* **14** (2025); DOI 10.1038/s41377-025-01755-8.
10. Synopsys Optical Solutions Group, *CODE V Reference Manual*, Release 11.7 (2024). [AUT, UMC user merit function.]
11. Eikonal Bridge Project, *PC1–PC11 Validation Atlas*, versions 1.x–2.x (2025–2026).
12. Eikonal Bridge Project, “Case A Manuscript — $S = F$ Identity Demonstration,” v2.1 (2026-04-21).
13. Eikonal Bridge Project, “Case C Manuscript — σ_W Scaling Span,” v6.0 (2026-04-21).
14. Eikonal Bridge Project, `LITERATURE_ANCHORS.md`, v1.1 (2026-04-21).
15. Eikonal Bridge Project, “Style Reference Template v1.0 (PC Revision Spec),” (2026-04-22).